S1 Supplementary Methods

Transition law, local geometry, and estimator validity boundaries

Article type: Supplement

Short title: S1 Methods - NDT

Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

1. Scope

This supplement records the mathematical objects named in the main text. It is not a claim that every object currently has a validated empirical estimator. NeuralManifoldDynamics implements selected chart-level functionals; it does not define NDT.

2. The transition law as the central object

The working process class is the controlled diffusion of the main text. The object that unifies drift, diffusion, reachability, committor, and control is the **control-dependent transition law**, not any one derived scalar.

The main text now carries the generator $\mathcal{L}_{t \mid c}$ and two related kernels: the latent law $K^{X \mid \gamma}_{t:t+h}$ and the chart-conditional law $K^{x \mid \gamma}_{t:t+h}$. This supplement records estimator and chart-dependence limits that do not belong in the narrative.

The snapshot kernel $K^{X,h}_c(\xi, A) = P(X_{t+h} \in A \mid X_t = \xi \text{ and } c_t = c)$ is a **special case** in which the protocol γ is constant on $[t : t + h]$. If c varies, or if history H_t matters, either augment the state $Z_t = (X_t, C_t, H_t, ...)$ until it is approximately Markov, or condition on the protocol path as in the main-text kernels. HELS-A and P6 are statements about γ, not violations of the controlled diffusion.

When the text writes K^h without a superscript, the intended object is the chart-conditional law K^x unless a latent argument is explicit. K^h is the law of the state at horizon h, not the path-space measure on $X_{[t:t+h]}$ or $x_{[t:t+h]}$. The family $\{K^h\}_{h>0}$ is the primary finite-horizon object. Committors are path/hitting functionals of the corresponding process (Klus et al., 2018; Metzner et al., 2009).

Relation to named summaries (same mapping as the main text):

- Drift b is the infinitesimal conditional mean rate.
- Diffusion a is the infinitesimal conditional covariance rate.
- The empirical Jacobian $\hat{J}_x$ is a chart-level derivative of $\hat{b}_x$; the latent counterpart is $\nabla_\xi b$.
- Stochastic short-horizon reachability is a finite-horizon predictive-spread functional of K^h.
- The committor is a hitting probability under the same process.
- Attractors organize the long-time law.
- FAR is the response of the law to a finite displacement.
- Rhythm, thalamus, history, body, and metabolism are candidate changes of K or $\mathcal{L}$, not extra theories.

NDT-C1–C3 below match the main text. RVC, thalamic access, HELS, EAP, and dendritic M_b are candidate mechanisms for C3. They are not part of the definition.

Three core commitments

NDT-C1 (reduced-state insufficiency). Current location in a reduced measurement chart x_t need not determine the latent future law K^X.

NDT-C2 (non-redundant accessibility). In at least some distinct neural regimes, functionals of the chart-conditional law provide reproducible out-of-sample information not contained in current chart location and standard static or spectral summaries.

NDT-C3 (conditional control). Context, history, or physiology can alter $K^x_h(\cdot \mid x, \gamma)$ among observations matched on current chart state.

3. Drift split and identifiability

Potential-driven drift supplies one structured component, $b = -M\nabla V + u_{\text{nc}}$, with mobility $M \succ 0$ (and $M = g^{-1}$ if a metric g is declared). NDT does not currently assume a fluctuation–dissipation relation linking M to a.

A residual non-conservative term is permitted because cyclic or asymmetric flow need not be gradient descent under the declared geometry. The split is **not** uniquely identified from trajectories without extra structure. An anisotropic or state-dependent M can make a gradient flow look non-normal in Euclidean coordinates. Chart Jacobian non-symmetry is therefore not proof of a genuine u_{nc} component. V is the NDT potential, not IIT's Φ.

4. Coordinate dependence

MNPS is an operational chart. Away from a fixed point, $J = \partial b/\partial x$ is not generally an intrinsic tensor under nonlinear reparametrization. An Itô SDE acquires an Itô correction in the transformed drift. Consequently α, ω, Ω, and local deformation summaries are **chart-level measurements** unless a connection is declared and a covariant derivative is used. NMD currently estimates the operational chart Jacobian. An intrinsic ∇b is not implemented and is not claimed.

5. Stochastic short-horizon reachability

The ellipsoid constructed by propagating local covariance under a locally linear stochastic model is **not** classical control-theoretic reachability and not a Gramian of a structural connectome. Call it **stochastic short-horizon reachability**: a covariance-based approximation to finite-horizon predictive spread under K^h. Horizon, residual quality, and neighbourhood leakage are validity conditions. NMD does not emit this object without an admissible one-step residual. Condition mixtures can also inflate the innovation term itself. In a companion Allen analysis, pooling expected and omission epochs increased $\mathrm{Tr}(Q)$ in all 126 examined session $\times$ region combinations, typically several-fold; residual degrees of freedom were 1 throughout those six-epoch fits of A, Q, and $\Sigma^{(h)}$. Condition-separated fitting is therefore required before any predictive-spread interpretation. Pearson coupling of regional chart-speed sequences is a different object and is not this ellipsoid.

6. Committor and finite-amplitude resilience

The committor $q_{AB}(x)$ is the probability of hitting B before A. It is destination geometry, not spread. Finite-sample 2-D recovery is not a current empirical export. FAR asks whether a finite kick can dislodge the state; it depends on perturbation family and amplitude and is not a local-linear summary. Neither object is licensed as an NMD v2 export in the present contract.

7. What this supplement does not authorize

Success at claim class 1 does not license $\mathcal{L}$ identification. Simulator envelopes for a, α, and ω are truth-known qualifications under restricted linear SDEs. They do not identify MNPS coordinates with latent X_t.

Bibliography

Klus, S., Nüske, F., Koltai, P., Wu, H., Kevrekidis, I., Schütte, C., & Noé, F. (2018). Data-driven model reduction and transfer operator approximation. *Journal of Nonlinear Science*, *28*, 985–1010. https://doi.org/10.1007/s00332-017-9437-7

Metzner, P., Schütte, C., & Vanden-Eijnden, E. (2009). Transition path theory for Markov jump processes. *Multiscale Modeling & Simulation*, *7*(3), 1192–1219. https://doi.org/10.1137/070699500